Pulley System Design

words and labels
numerical
symbolic eq.
Important info & answers

knowns

Pulley dia. = 3 in center distance = 4 in

Rotational speed = 1050 - 2050 rpm (1800 rpm)

Static belt tension = 4 - 6 lbf (4 lbf)

pulley material = Aluminum, ABS, steel

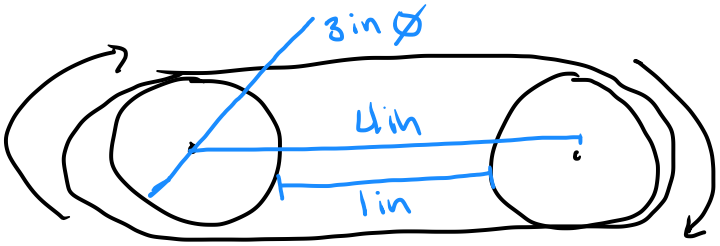
Safety factor = 3

Belt material = polyurethane rubber

↳ yield strength = 3.25 MPA / 471 psi at 20% strain

unknowns

belt length
ω rad/sec
torque
power
belt diameter



total belt length: Curved sections + Straight sections

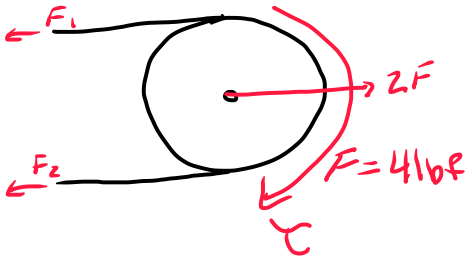
$3\pi + 8 = 17.42$ in



Rotational speed → linear speed

$1800 \frac{rev}{min} \cdot \frac{3\pi\ in}{rev} \cdot \frac{1\ min}{60\ sec} = 282.74 \frac{in}{sec}$

$1800 \frac{rev}{min} \cdot \frac{2\pi\ rad}{rev} \cdot \frac{1\ min}{60\ sec} = 188.50 \frac{rad}{sec}$



$F_1 > F_2$

$F_1 + F_2 = 2F = 8$ lbf

$P = \omega r (F_1 - F_2)$

$\tau = r(F_1 - F_2)$

$\frac{RPM \cdot \pi D}{60} = V$

$RPM \cdot \frac{2\pi}{60} = \omega$

$\mu = 0.60$ According to GORTEF between Rubber and steel in a kinetic setting

$\tau = 1.5\text{in}(6.9454 - 1.0546)$

$\tau = 8.8362$ in-lbf

$\frac{F_1}{F_2} = e^{\mu\theta} = e^{0.6 \cdot \pi\ rad} = 5.724$

$\tau \cdot 188.5 \frac{rad}{sec}$

↳ $= 1665.58 \frac{in\text{-}lbf}{sec}$

$F_1 = 8 - F_2 \rightarrow \frac{8 - F_2}{F_2} = 6.586$

$\frac{8}{F_2} - 1 = 6.586$ $F_1 = 6.9454$

$F_2 = \frac{8}{7.586} = 1.0546$ lbf



$$S_{allowable} = \frac{S_y}{S_F} = S_{MAX}$$

$$S_{MAX} = \frac{471}{3} = 157 \text{ psi}$$

$$S_{max} = \frac{F_{max}}{A} \rightarrow A = \frac{F_{max}}{S_{max}}$$

$$A = \frac{6.9454 \text{ lbf}}{157 \text{ psi}} = .04424 \text{ in}^2$$

Belt selection

Polyurethane Rubber

D > d →

flexible

Circumference > 17.42 in

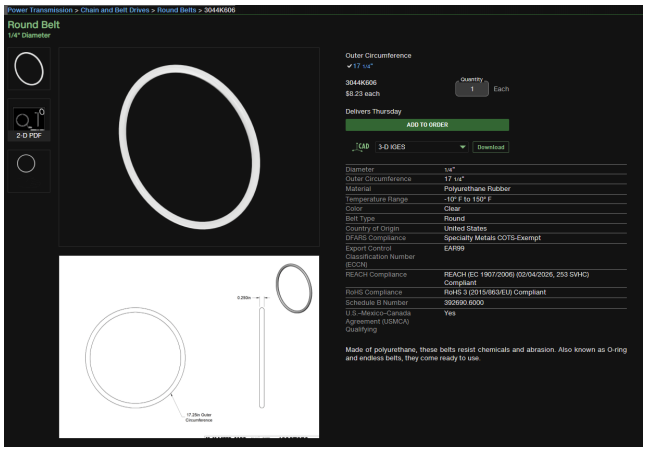
$$A = \pi \frac{d^2}{4} \qquad d = \sqrt{\frac{A \cdot 4}{\pi}}$$

$$d = 0.23783 \text{ in}$$

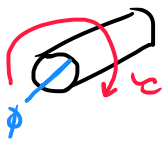
McMaster-Carr

product ID: 3044K606

1/4 in dia, chosen circumference range contains our value



Shaft design



$$S_{shear} = \frac{S_y}{2 \cdot S_F} \qquad S_{MAX} = \frac{93000 \text{ psi}}{2 \cdot 3}$$

$$= 15\,500 \text{ psi}$$

knowns

material = 1060 steel

↳ 93000 psi yield strength

Pulley dia. = 3 in

Belt tensions = F_1 and F_2

Torque = 8.8362 in-lbf

Safety factor = 3

$$S_{MAX} = \frac{\tau \cdot \frac{d}{2}}{J}$$

$$J = \frac{\pi d^4}{32}$$

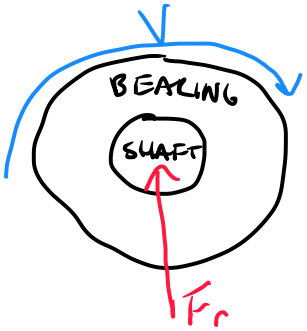
$$S_{MAX} = \frac{\tau \cdot 16}{d^3}$$

$$d = \sqrt[3]{\frac{\tau \cdot 16}{\pi S_{max}}}$$

$$d = \sqrt[3]{\frac{8.8362 \cdot 16}{\pi \cdot 15\,500 \text{ psi}}}$$

$$d = 0.1427 \text{ in}$$

Bearing Design



knowns

shaft dia.
pulley dia
radially Force = 2F
RPM

Unknowns

Bearing pressure
Bearing velocity
PV rating
material

Bearing Length

↳ 0.25 in
chose this in prev. assignment.

$$P = \frac{F_r}{L \cdot d}$$

$$= \frac{8 lbf}{0.25 \cdot 0.1427}$$

$$= 224.2 \text{ psi}$$

$$V = \frac{RPM \cdot \pi d}{12} = Ft/min$$

$$\frac{1800 \cdot \pi \cdot .1427}{12}$$

$$= 67.2 \text{ ft/min}$$

$$PV = P \cdot V \cdot SF$$

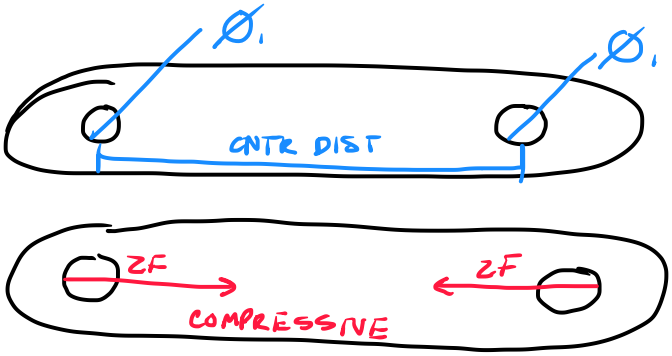
$$= 45289$$

chosen material:
Porous Bronze

PV = 50000
P = 4500 V = 1500

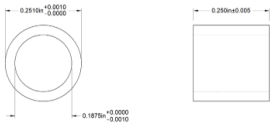
Frame Design

OD of Bearing - interference



I'll just make the frame as thick as the bearing.

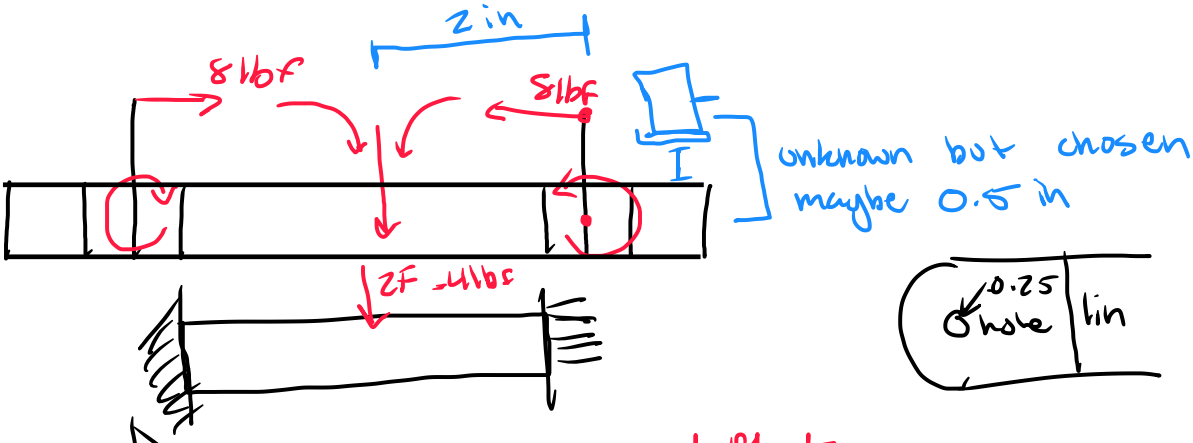
I'll use FEA to guess and check for a suitable design



closest shaft dia is 3/16 so I found a bronze bearing for that

hole dia probably 0.250 so it will have interference.

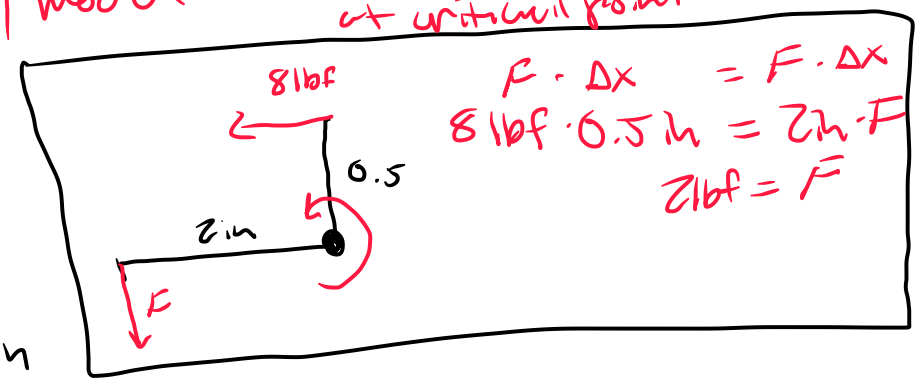
this means interference range is between 1 and 2 thou - FN5 fit





model

Max deflection at critical point



$F \cdot \Delta x = F \cdot \Delta x$

$8\,lbf \cdot 0.5\,in = 2\,in \cdot F$

$2\,lbf = F$

MAX deflection at load

$\frac{WL^3}{192EI}$

$L = 3.75$ in

$W = 4$ lbf

E = material = aluminum 6061 = 10000 ksi

↳ familiar material

$I = \frac{bd^3}{12}$ $\quad b = 1$ in $\quad d$ = unknown thickness

$\delta_{MAX} = 0.005$ in

↑ previous assignment guesstimate

$\frac{\delta_{MAX}}{5F} = \frac{WL^3 \cdot 12}{192 E b d^3}$ $\quad d = \sqrt[3]{\frac{WL^3 \cdot 12 \cdot 5F}{192 \cdot E \cdot b \cdot \delta_{max}}}$

$$\sqrt[3]{\frac{4 \cdot 3.75^3 \cdot 12 \cdot 3}{192 \cdot 10000 \cdot 10^3 \cdot .005}} = 0.0924829527874$$

thickness can just be 0.25 in because its the length of the bearing.

